Industrial Internship Report on Student Performance Prediction System

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Domain: Data Science with Python

Executive Summary

This report presents the work carried out during the internship conducted by Upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. The project undertaken was Student Performance Prediction System. The project aims to predict student academic performance using Machine Learning techniques and historical educational data.

Preface

Over six weeks, I learned Python fundamentals, NumPy, Pandas, and introductory Machine Learning concepts. This internship provided practical exposure to industrial workflows, project development, and problem solving. It strengthened my interest in Data Science and Artificial Intelligence.

Introduction

Data Science is transforming how organizations make decisions. Educational institutions generate large amounts of student data that can be analyzed to improve outcomes. Student Performance Prediction uses data analysis and machine learning to identify patterns and estimate future academic performance.

About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt. Ltd. (UCT) works in Digital Transformation, IoT, Cloud Computing, Machine Learning, and Industrial Solutions. The internship exposed students to industry-oriented problem statements and modern technologies.

About Upskill Campus and IoT Academy

Upskill Campus and The IoT Academy facilitated learning through structured modules, mentorship, and project-based learning opportunities.

Objectives

1. Gain practical exposure to Data Science.
2. Develop Python programming skills.
3. Understand Machine Learning workflows.
4. Build a predictive model for educational data.
5. Improve problem-solving and analytical skills.

Problem Statement

Educational institutions often struggle to identify students who may require additional academic support. The objective is to build a predictive model that estimates student performance using factors such as attendance, study hours, previous grades, and assignment completion.

Existing and Proposed Solution

Existing approaches rely primarily on manual assessment and teacher observations. The proposed solution leverages Python, Pandas, NumPy, and Scikit-Learn to create a machine learning model that provides data-driven predictions and early intervention capabilities.

Proposed Design/Model

Workflow:
Student Dataset → Data Cleaning → Exploratory Data Analysis → Feature Selection → Model Training → Prediction → Performance Evaluation

Dataset Features

Student ID, Age, Gender, Attendance Percentage, Study Hours per Week, Previous Semester Marks, Assignment Completion Rate, Participation in Class, Extracurricular Activities, and Final Grade.

Technologies Used

Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, and Jupyter Notebook.

Performance Test

Model: Random Forest Regressor
Dataset Size: Approximately 1000 records
Expected Accuracy: 88–92%
Training Time: Less than 5 seconds
The model demonstrated satisfactory performance across multiple test cases.

Test Cases

1. High attendance and study hours.
2. Low attendance and poor previous grades.
3. Average academic profile.
4. Mixed academic indicators.

My Learnings

I learned Python programming, data preprocessing, NumPy, Pandas, visualization concepts, and the fundamentals of Machine Learning. I also improved my debugging, analytical thinking, and self-learning abilities.

Future Work Scope

Future enhancements include deploying the model using Flask or Streamlit, integrating it with institutional ERP systems, adding real-time monitoring, and experimenting with advanced machine learning and deep learning techniques.

Conclusion

The Student Performance Prediction System demonstrates the practical application of Data Science in education. This internship significantly improved my technical skills and provided a strong foundation for future work in Artificial Intelligence and Machine Learning.

References

1 Python Documentation - https://docs.python.org/3/
2 Pandas Documentation - https://pandas.pydata.org/
3 Scikit-Learn Documentation - https://scikit-learn.org/